DATABRICKS · HEALTHCARE / ICU OPERATIONS

ICU Step-Down Readiness

Turning scarce ICU capacity into a decision advantage — an AI-assisted way to see, every shift, which patients are most ready to safely leave intensive care.

Lakeflow | Unity Catalog | Lakebase | Mosaic AI + Foundation Models | Genie | Databricks Apps

THE PROBLEM

A blocked ICU bed is a clinical and financial bottleneck

70–90%

Typical ICU occupancy — one blocked bed delays a critical admission

Manual

Step-down decisions are subjective and vary by provider and shift

"To be safe"

Clinicians hold patients longer than needed, lengthening stays

The question every intensivist asks on rounds — *“who can safely move to step-down today?”* — has no consistent, data-driven answer.

CUSTOMER STORY · NORTHBRIDGE REGIONAL (ILLUSTRATIVE)

7 a.m.: a full ICU, and two critical patients waiting in the ED

A TYPICAL MORNING TODAY

7:00	Rounds start at bed 1; all 24 patients reviewed in bed order
11:30	First step-down call made; one stable patient kept another day “to be safe”
14:00	Beds finally free; ED patients have boarded for 7 hours
15:00	Elective cardiac case postponed for lack of a post-op ICU bed

THE SAME MORNING WITH THE READINESS INDEX

6:45	Charge nurse opens the census already ranked; the team starts rounds with the five most-ready patients
8:15	Beds 7 and 12 confirmed ready: one off pressors for 36 hours, one extubated and stable. Bed 3 ranks high but was ventilated overnight, so it's capped at Borderline and stays
10:00	Both transfers complete; the two ED patients are in ICU beds by 11:00
Next day	Elective case goes ahead; neither transferred patient bounces back

> “It didn't make the call for us. It told us where to start, and why.”
> ICU medical director, Northbridge Regional

Illustrative composite scenario: Northbridge Regional and all patients are fictional. The gains shown are what the pilot is designed to measure, not reported results.

WHY IT MATTERS BEYOND THE UNIT

ICU length of stay is a margin, benchmarking, and market-share lever

Payer & margin risk

Under DRG-fixed and value-based contracts, every avoidable ICU day is **unreimbursed cost**, and ICU bounce-backs feed readmission penalties and quality scores (CMS HRRP, star ratings) that payers price on.

Peer benchmarking

ICU LOS, occupancy, and throughput are ranked against peer hospitals (Vizient, CMS, Leapfrog). Being an **LOS outlier** is a board- and payer-visible signal of inefficiency.

Competitive position & growth

A full ICU means **diverted ambulances, declined tertiary referrals, and postponed elective surgery** — lost revenue, lost referral relationships, and lost market share. Freed capacity is growth capacity.

The intensivist's daily step-down call, aggregated across the unit, is what a **CFO, CMO, and market strategist** see as reimbursement exposure, a benchmark percentile, and the ability to say yes to the next transfer.

THE OUTCOME WE SELL

Free ICU capacity without compromising safety

10–15%
Target reduction in ICU length of stay

$2–5k
Saved per accelerated discharge-day

+beds
More ICU capacity for emergent admissions

Consistency
Evidence-informed decisions across shifts

For a 30-bed ICU, a 0.8-day average LOS reduction is on the order of **hundreds of additional patient-days of capacity per year**.

WHAT WE BUILT

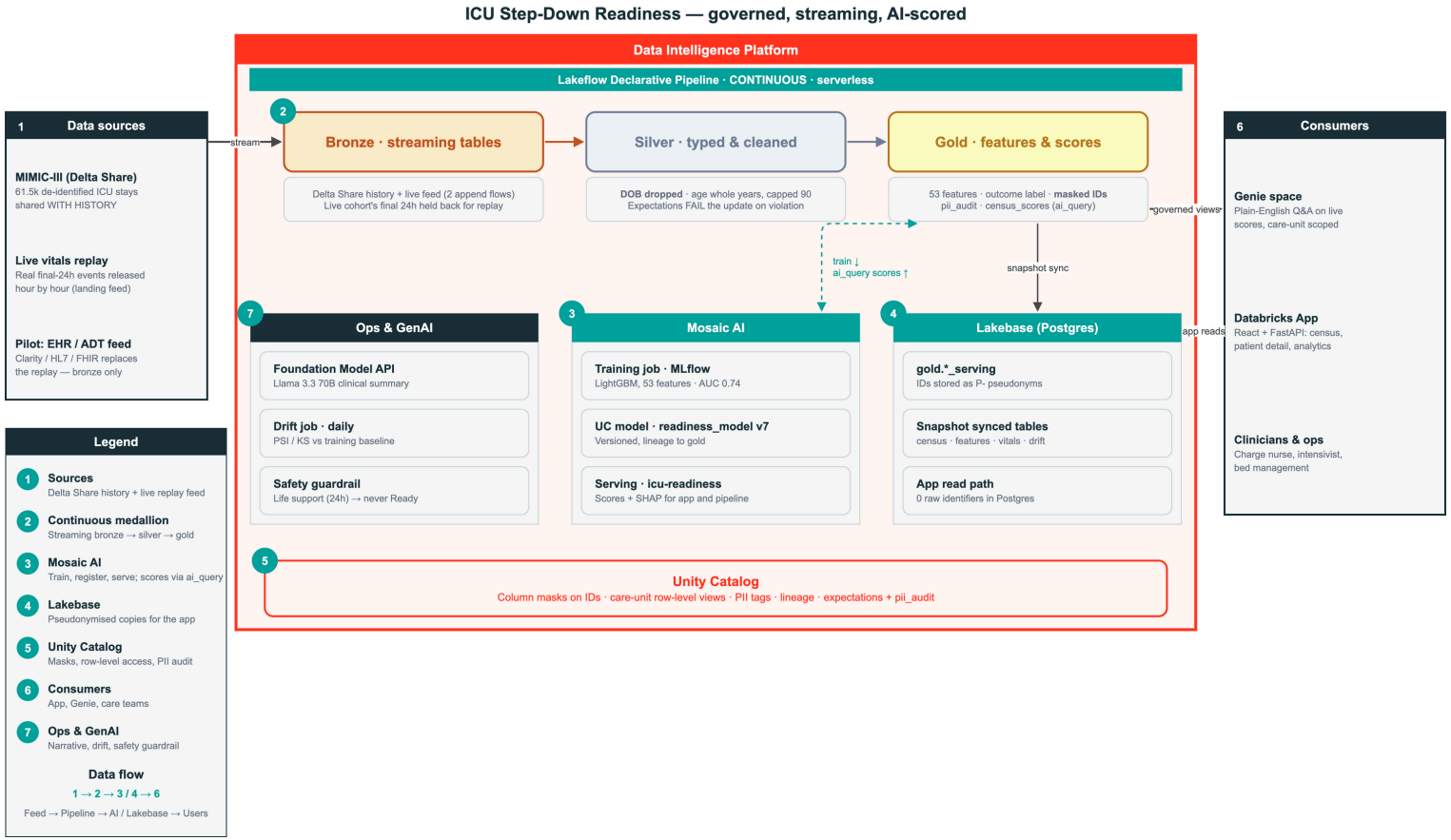
One governed journey — raw data to a clinician's screen

MIMIC-III (Delta Share) → **Lakeflow medallion** → **Unity Catalog** →
Lakebase serving → **ML + Gen AI** → **Genie** → **Databricks App**

Integrated, not siloed: the same governed gold tables feed the operational app, the model, and natural-language querying — one lineage, one source of truth.

ARCHITECTURE

Governed, continuously streaming, AI-scored — on one platform



Source data streams in continuously; silver drops date of birth; gold carries masked identifiers and live model scores; the app reads pseudonymised Lakebase copies; Unity Catalog governs every layer. Editable source: `deck/architecture.drawio`.

THE PRODUCT

A ranked ICU census the care team acts on

- **Census dashboard** — every current patient ranked by a relative readiness index (re-scored within minutes as new vitals stream in), color-banded Ready / Borderline / Not-ready.
- **Patient detail** — the factors driving each recommendation, vital-sign trends, and a plain-language **AI clinical summary**.
- **Analytics** — capacity and outcome trends, with honest model performance (v7: AUC 0.74, 0.64 readmission-only).

Live: `icu-step-down-7474645901826067.aws.databricksapps.com`

Example — ranked census

🟢 **92** · Ready · off pressors 36h, GCS 15

🟢 **78** · Ready · extubated, stable VS

🟡 **54** · Borderline · lactate ↑

🔴 **18** · Not ready · on vent + pressors

FOR THE CMIO & ICU MEDICAL DIRECTOR

Built to be trusted, governed, and operated

Governed by design

Unity Catalog across the medallion — DOB dropped, identifiers pseudonymized via UC column masks, PII classification tags, least-privilege grants, full bronze→gold lineage with continuous audit.

Live-fed & operationally served

Continuous Lakeflow streaming pipeline — new vitals in bronze in seconds, scores in ~2 min; pseudonymised gold synced to Lakebase (Postgres) for low-latency app reads; model served on Mosaic AI; narrative via Foundation Model API.

Reproducible

Everything deploys via Databricks Asset Bundles; 199 backend + 40 frontend + 74 pipeline/ML unit + 28 live integration tests, run output committed.

Safe by framing

Public de-identified MIMIC; a relative decision-support index, not an automated discharge gate.

MODEL HONESTY

Decision support, not a discharge switch

AUC 0.74

95% CI 0.71–0.76 on patients who left the ICU alive — the only patients a step-down decision is made for. Flags not-safe step-downs at ~5× the base rate

Readmission-only 0.64

72h bounce-back alone is genuinely hard from structured data. We say so, rather than hide it

Index, not %

Relative readiness rank within the current census — never a "probability of safe discharge"

Explainable: SHAP factors in clinical language; known risks always shown as risks, and a patient on vasopressors or a ventilator in the last 24h is never banded Ready. We retired an earlier 0.89: it counted patients who died in the ICU, whose final day is easy to recognise. The next lift comes from your data, notes and validation with your clinicians.

PROJECT PLAN

Trained on your history, proven on your unit — before anyone relies on it

Weeks 1–4 · Train on your … | Weeks 5–8 · Silent pilot | Weeks 9–16 · Clinician pilot | Week 17+ · Scale

PHASE 1 · WEEKS 1–4

Train on your historical data

- Point the same governed pipeline at your EHR feed — 2–3 years of your ICU stays
- Retrain on your patients, practice and outcomes
- ICU Medical Director co-defines features, guardrails and the “safe step-down” label

Exit gate: discrimination and calibration on your held-out stays, checked by unit and patient subgroup

PHASE 2 · WEEKS 5–8

Silent pilot

- Live scores every shift, *not shown* to clinicians
- Compared with the step-down calls clinicians actually made, and 72h outcomes
- Validates the numbers on today's patients, not just history

Exit gate: live performance matches retrospective; no safety signal in Ready-banded patients

PHASE 3 · WEEKS 9–16

Clinician pilot — one ICU

- Ranked census used on rounds; the decision stays with the clinician
- Clinicians record agree / disagree and why
- Weekly review of flagged vs actual bounce-backs builds trust and tunes thresholds

Exit gate: ICU LOS reduced with safe-transfer and 72h bounce-back rates held or improved

PHASE 4 · WEEK 17+

Decide & scale

- Go / no-go with clinical governance
- Extend unit by unit on the same platform
- Drift monitoring and scheduled retraining

Ongoing: performance reported to the clinical governance board

Clinicians see nothing until the model has been validated twice: first on your history, then silently on live patients. Trust is earned before the index is used on rounds.

THE ASK

From prototype to a unit pilot

- **VP of Clinical Operations / CMO** — measured on ICU length of stay, bed occupancy, and emergent-admission wait times: sponsor the plan, with an 8-week clinician pilot in one ICU scored on LOS reduction with the safe-transfer rate held or improved.
- **ICU Medical Director** — measured on safe-transfer rate and 48–72h ICU bounce-back: co-define the readiness features and validate the index against your step-down criteria.
- **CMIO / VP Data & Analytics** — measured on time-to-value and governed reuse: point the same pipeline at your EHR feed — the architecture is unchanged.

Every ask maps to a number already on that leader's own scorecard. The capability is live today on Databricks; the next step is your data and your clinicians.